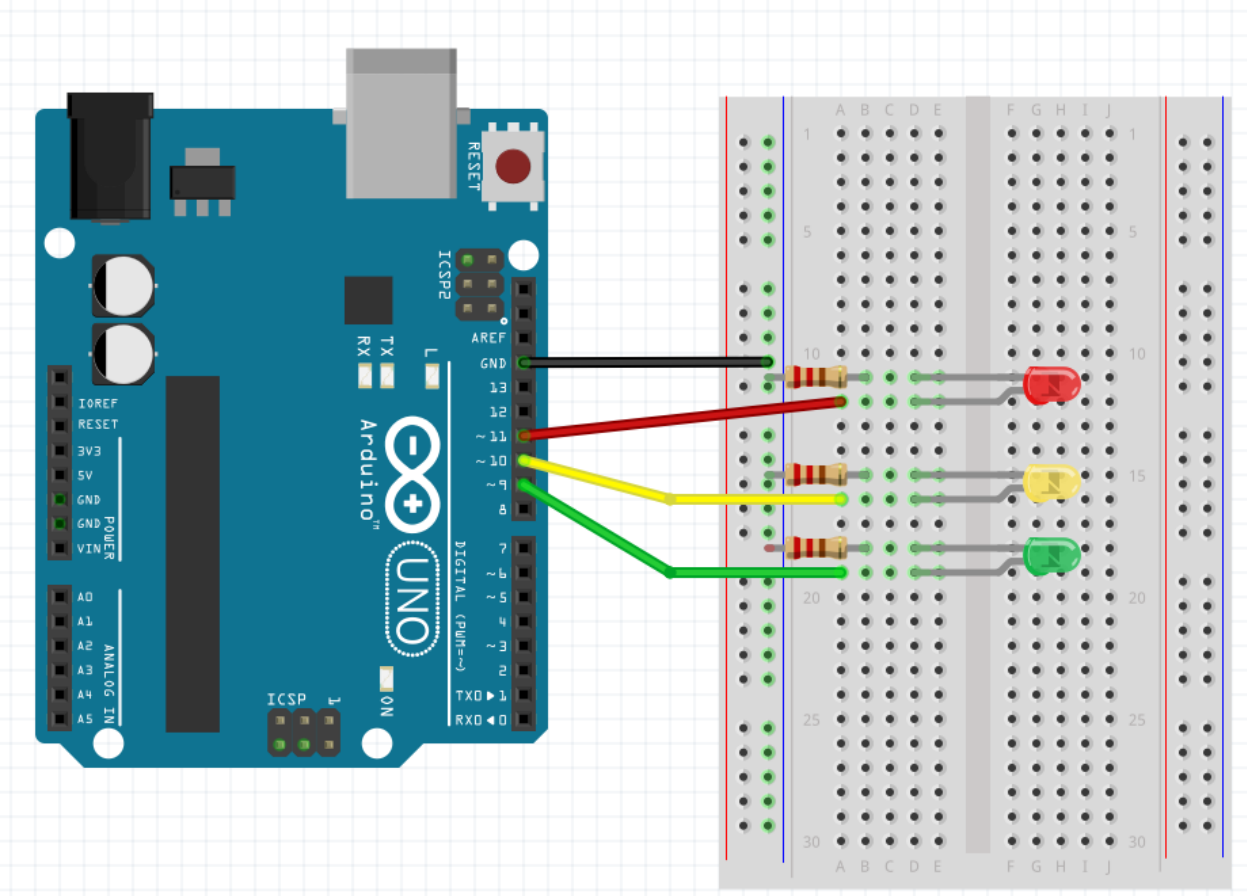
Traffic Light Assignment

Purpose: To wire and code a series of lights to operate a traffic light signal.



Materials: Computer, Arduino Uno Board, USB cable, Breadboard, 1 green LED, 1 yellow LED, 1 red LED, 3 - 220 Ω resistors.

New Components:
There are no new components, just more of them!

Procedure:

1. Wire the project using the following schematic.

2. You can create your code in TinkerCAD

3. Below are the blocks that you will need, click on the Arduino button to convert the blocks into Arduino code.




```
void setup()
{
  pinMode(9, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(11, OUTPUT);
}

void loop()
{
  digitalWrite(9, HIGH);
  delay(7000); // Wait for 7000 millisecond(s)
  digitalWrite(9, LOW);
  digitalWrite(10, HIGH);
  delay(3000); // Wait for 3000 millisecond(s)
  digitalWrite(10, LOW);
  digitalWrite(11, HIGH);
  delay(10000); // Wait for 10000 millisecond(s)
  digitalWrite(11, LOW);
}
```

*****SAVE your sketch to your Robotics/Coding Folder that you created*****

5. Copy the Arduino Code into the TinkerCAD website. To paste the code into the website be sure to use ctrl “v.” Hook up your Arduino to your computer using the USB cable.

6. Upload your program to the Arduino. Make sure the Arduino Uno board is hooked up using the supplied USB cord to your computer and click the green “upload” button.

7. Your program is now loaded. Observe your LEDs and complete the questions and EXTENSION PROJECTS on your worksheet titled Assignment: Traffic Light.

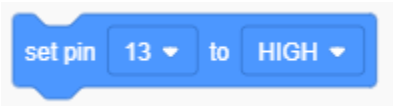
Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the leds, is the long positive arm connected by wires to the pins?

c. It may be wise to test all three leds by turning them all on. Open a new Arduino window and create this test code. The blocks should look like this:



d. Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when you try this test code? If no



e. Next check the individual wires in the circuit. This can be done by connecting the LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.

f. Check the button. This can be achieved by connecting it and using the serial monitor built into tinkerCAD. Load this test program:



As you hold the button down the serial monitor should change to on from off.

Extension Challenge:

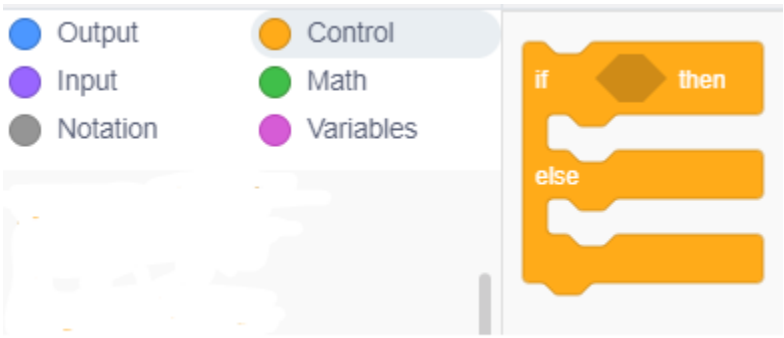
For an extension, let's add either a **button** or a **potentiometer**, and use them to control how long each light is displayed for:

1. Begin by changing the current timings so that *each light stays on for only 1 second.*

2. Wire in a new circuit to allow checking if a button is pressed or the angle of the potentiometer's rotation. *(See previous activities instructions to remember how to do*

this)

3. If either the button is pressed or the potentiometer's analogRead() value is less than 500, have the timings be shorter *so that each light stays on for only 0.5 seconds.*

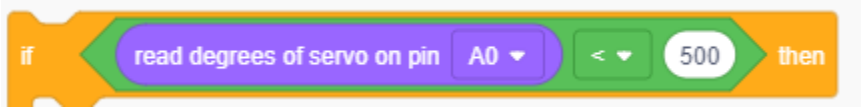


4. To do this, you'll need to add some logic. Use the IF - ELSE block in the **Control** section.

5. The *condition* for the IF statement should depend on what component you are using. Either:



or



6. You can place a copy of your existing code into each branch of the IF-ELSE, and change the delay timings accordingly.

Because the delay block halts the program for a certain amount of time, any checks to see where a potentiometer is or if a button is pressed can't happen during this time. With the original timing, the code would only check for these events once every 20 seconds! The revised version allows the check to happen at least once every 3 seconds; although this still isn't great, it is better. Shortly we'll look at an improved coding technique so that timings don't stop the rest of the program from continuing.

Assignment: Traffic Light.

1. How long do each of the lights stay on?

2. What part of the code dictates how long the light stays on.

3. How would you change the timing of the lights to accommodate a very busy road. For example, you want the traffic on this street to move for a longer period of time than it is stopped. Code your lights to be green for 4 seconds, yellow for 0.5 seconds and red for 2 seconds.